HEALTH INSURANCE COST PREDICTION USING MACHINE LEARNING

PROJECT REVIEW

SET ID - 223714

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INTRODUCTION

We live on a planet full of threats and uncertainty. Including People, households, durables, properties are exposed to different risks and the risk levels can vary. But, risks cannot usually be avoided, so the world of finance has developed numerous products to shield individuals and organizations. Therefore Insurance is one of the policies that either decreases or removes loss costs incurred by various risks. The value of insurance in the lives of individuals.

So, this project focuses on predicting health insurance coverage using Machine Learning techniques. Early estimation of health insurance costs can help individuals consider the required amount more thoughtfully.

OBJECTIVES

1. Develop machine learning models for accurate health insurance cost prediction.

2. Identify high-risk individuals and implement risk mitigation strategies.

3. Create a user-friendly web application using Streamlit for personalized cost estimation and interactive visualization

METHODOLOGY

DATASET:

I used an online dataset which is publicly available on the Kaggle website. It provides persons' information which includes over 1338 records and 7 attributes.

DATA PRE-PROCESSING:

I performed several pre-processing steps which are:

- **Handling Missing Values:** Identified and addressed missing data through imputation or deletion while ensuring data integrity.
- **Data Processing:** Encoded categorical variables, handled outliers, and converted data types for model compatibility.
- **Correlation Matrix:** Utilized a correlation matrix to understand relationships between variables and mitigate multicollinearity issues.

CLASSIFICATION TECHNIQUES:

- **I used different algorithms like: 1. Logistic Regression , 2. Gradient boosting, 3.Svm (Support Vector Machine) , 4. Random Forest. From the above algorithms we used Gradient Boosting for our model.**

- **Gradient boosting: It works by combining multiple weak learners sequentially, each correcting errors made by its predecessors. It minimizes a loss function by iteratively fitting new models to the residuals of the previous ones. Trees are commonly used as base learners, and the final prediction is the sum of all individual predictions.**

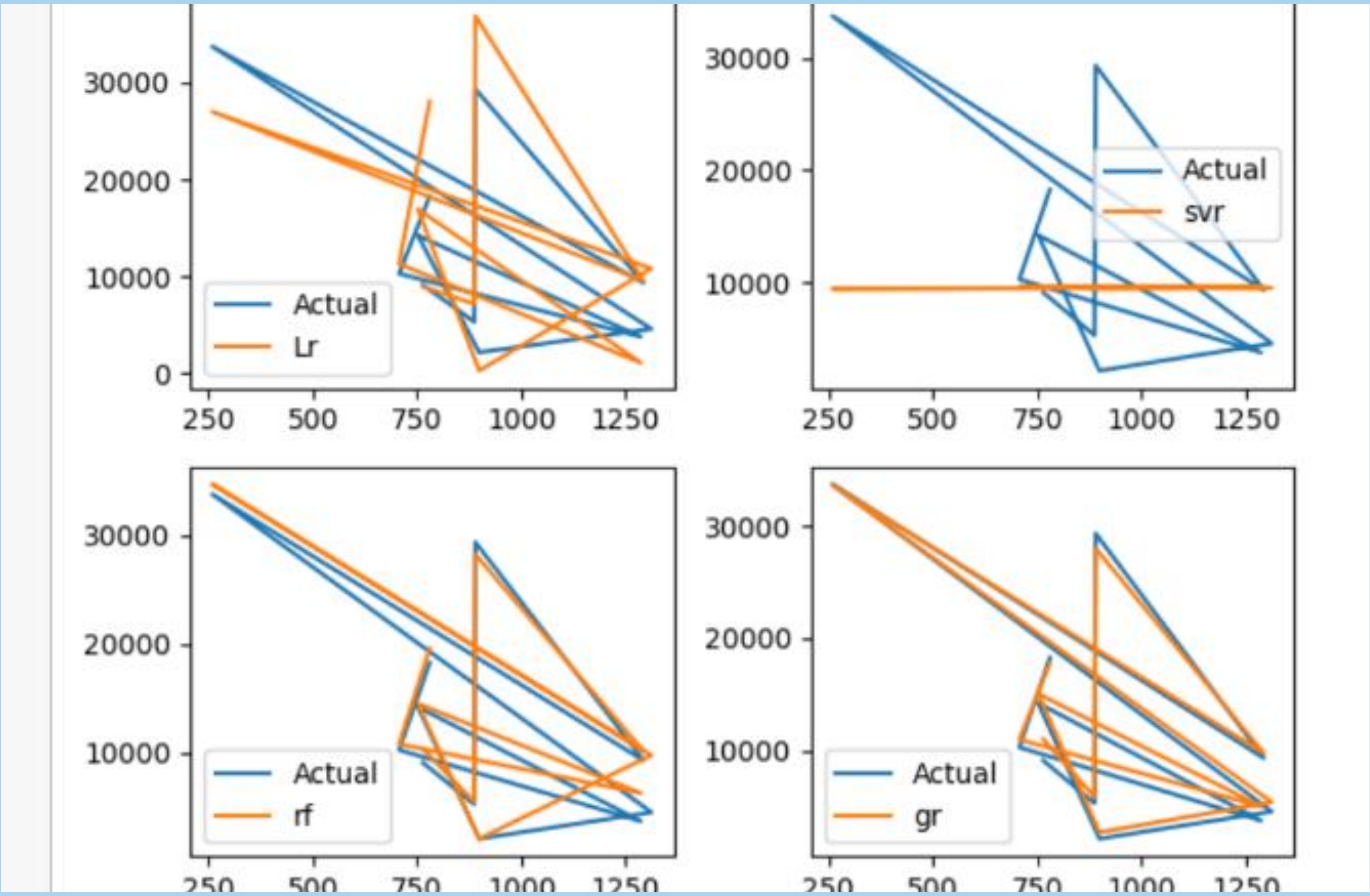
- **In health insurance cost prediction, gradient boosting can be used to model complex relationships between factors such as age, medical history, and coverage type. By training on historical data, it can accurately predict insurance costs for individuals based on their unique characteristics, helping insurers assess risk and set premiums effectively.**

ACCURACY AND PREDICTION:

After performing above algorithms we obtained different accuracy level for each algorithm which is listed in the table below.

	Models	ACC
0	LR	0.783346
1	SVM	-0.072298
2	RF	0.865756
3	GB	0.877973

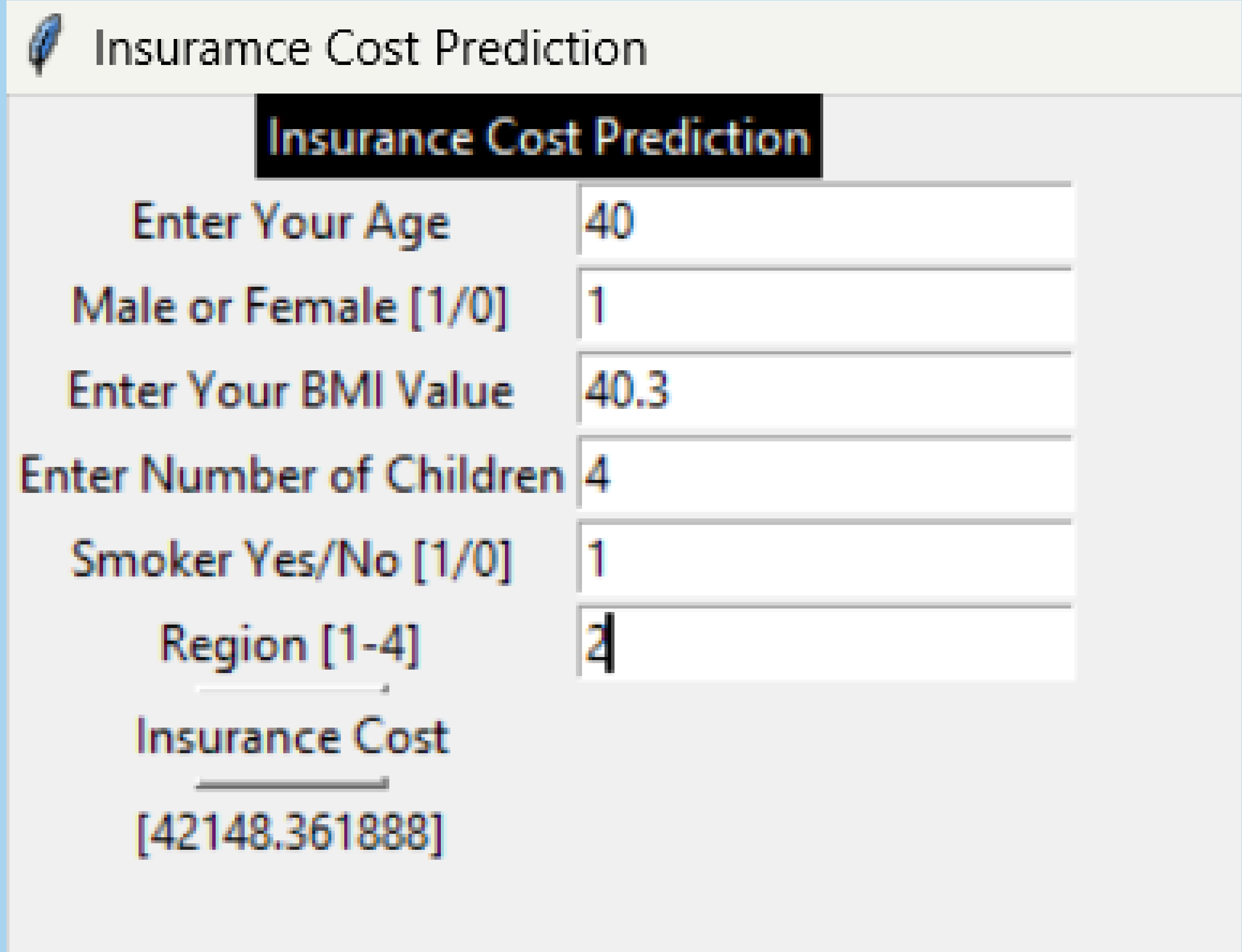
Based on the table provided, we can conclude that Gradient Boosting is the suitable method which can be used to develop our model. Also, I plotted these algorithm to compare the performance visually.



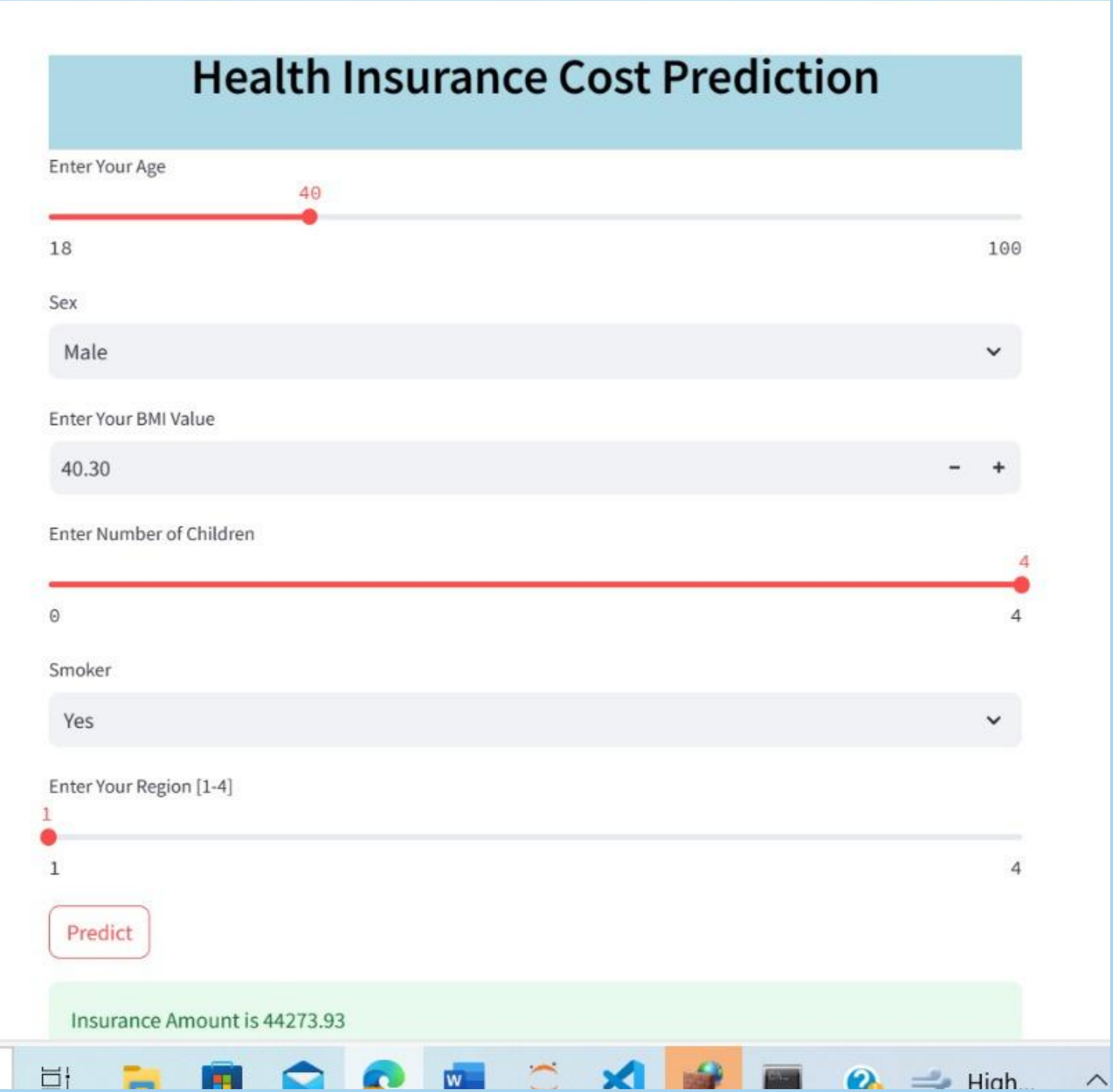
It's evident that the Gradient Boosting algorithm achieves the highest level of accuracy. So, I used Gradient Boosting algorithm to develop our model.

MODEL BUILDING:

I developed a Python script that creates a basic graphical user interface (GUI) application to predict Insurance's cost using Tkinter. It creates a window where users can enter various medical parameters, which are then used to make predictions using a pre trained machine learning model.



It's recommended to enrich this code by adding more effective error running and ensuring that the template file is available before loading it.



I developed a web page using Streamlit, this Streamlit web page offers users a seamless experience in predicting insurance costs, providing accurate estimates based on customizable input variables. With intuitive design and real-time feedback, it simplifies the process of understanding potential insurance expenses.

LITERATURE REVIEW

TITLE: Claim Churn Prediction (2019) **AUTHOR:** Ranjodh Singh **FINDINGS:** The system utilizes image analysis of damaged vehicles to assess repair costs, aiding in the evaluation of claim expenses and facilitating informed decisions regarding insurance claims.	**TITLE:**Prediction of healthcare insurance costs(2021) **AUTHOR:**Ishiwata **FINDINGS:**This study investigates the use of machine learning for predicting healthcare insurance costs in Japan. It compares applying computational intelligence and Spark,for cost analysis.
TITLE: Machine Learning for Insurance Rate Making with Big Data (2017) **AUTHOR:**Sun et al **FINDINGS:**This work explores the application of machine learning algorithms to big insurance datasets for rate making in the insurance industry	**TITLE:** Deep Learning for Insurance Fraud Detection and Cost Prediction (2022) **AUTHOR:**Zhang **FINDINGS:**This study investigates the use of deep learning models to detect fraudulent insurance claims and predict insurance costs simultaneously.

CONCLUSION

This project provides the deep insight into machine learning techniques for predicting insurance cost. By predicting insurance rates based on a variety of factors, insurance policy firms may attract consumers and save time. Thus, Machine learning may significantly reduce these individual efforts in price analysis since ML models can compute costs quickly while doing so would take a person a long time.

REFERENCE

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Thank you!